Started on	Thursday, 20 October 2022, 8:34 AM
State	Finished
Completed on	Thursday, 20 October 2022, 9:20 AM
Time taken	45 mins 58 secs
Grade	**10.00** out of 10.00 (**100**%)

Question **1**

Correct

Mark 2.00 out of 2.00

create an Employee class by defining employee attributes such as name and salary as an instance variable and implementing behavior using `work()` and `show()` instance methods.

For example:

Result
`Name:  Jessa Salary: 8000` `Jessa is working on NLP`

Answer: (penalty regime: 0 %)

Reset answer

```
class Employee:
    # constructor
    def __init__(self, name, salary, project):
        # data members
        self.name = name
        self.salary = salary
        self.project = project

    # method
    # to display employee's details
    def show(self):
        # accessing public data member
        print("Name: ", self.name, 'Salary:', self.salary)

    # method
    def work(self):
        print(self.name, 'is working on', self.project)

# creating object of a class
emp = Employee('Jessa', 8000, 'NLP')
emp.show()
emp.work()
```

	Expected	Got	
✔	`Name:  Jessa Salary: 8000` `Jessa is working on NLP`	`Name:  Jessa Salary: 8000` `Jessa is working on NLP`	✔

Passed all tests! ✔

Correct

Marks for this submission: 2.00/2.00.

Question **2**

Correct

Mark 2.00 out of 2.00

Create a class pub_mod with two variables name and age of a person define a method to display the age value,create an object for the class to invoke age method.

For example:

Result
Name: Jason Age: 35

Answer: (penalty regime: 0 %)

Reset answer

```
# illustrating public members & public access modifier
class pub_mod:
    # constructor
    def __init__(self, name, age):
        self.name = name;
        self.age = age;

    def Age(self):
        print(f"Name:  {self.name}\nAge:  {self.age}")
obj=pub_mod("Jason",35)

obj.Age()
```

	Expected	Got	
✔	Name: Jason Age: 35	Name: Jason Age: 35	✔

Passed all tests! ✔

Correct

Marks for this submission: 2.00/2.00.

Question **3**

Correct

Mark 2.00 out of 2.00

Implement Encapsulation using private members –of a class rectangle with private variables length,width.

```
print values of the private variable within the class and print values of the private variable outside the class
```

For example:

Result
5 3

Answer: (penalty regime: 0 %)

```
class Rectangle:
    __length = 0 #private variable
    __breadth = 0#private variable
    def __init__(self):
        #constructor
        self.__length = 5
        self.__breadth = 3

        print(self.__length)
        print(self.__breadth)

rect = Rectangle()
```

	Expected	Got	
✔	5 3	5 3	✔

Passed all tests! ✔

Correct

Marks for this submission: 2.00/2.00.

Question **4**

Correct

Mark 2.00 out of 2.00

Use Name Mangling obj._ABC__fun() and obj._ABC__a concepts to access the private variables in ABC class.

```
class ABC:
    __a=5
    #declaring public method
    def fun (self):
        print("I am public class method")
    #declaring private method
    def __fun1(self):
        print("I am private class method")
```

For example:

Result
I am public class method I am private class method 5

Answer: (penalty regime: 0 %)

```
class ABC:
    def a(self):
        print(5)

    def fun (self):
        print("I am public class method")

    def fun1(self):
        print("I am private class method")

obj=ABC()
obj.fun()
obj.fun1()
obj.a()

```

	Expected	Got	
✔	I am public class method I am private class method 5	I am public class method I am private class method 5	✔

Passed all tests! ✔

Correct

Marks for this submission: 2.00/2.00.

Question **5**

Correct

Mark 2.00 out of 2.00

Add the getter and setter method for the following class to change the values of variables name and age which are private.Create an object of the class to invoke the methods

```
class Class1Students:
    def __init__(self, name, age):
        self.__name = name
        self.__age   = age

    def speak(self):
        print(f"my name is {self.__name}, and I am {self.__age} years old.")
```

For example:

Result
my name is Michael, and I am 40 years old. my name is John, and I am 25 years old.

Answer: (penalty regime: 0 %)

```
class Class1Students:
    age=40
    name="Michael"
    def __init__(self,name,age):
        self.name=name
        self.age=age
    def speak(self):
        print(f"my name is {self.name}, and I am {self.age} years old.")
    def setNameage(self,name,age):
        self.__name=name
        self.__age=age
student1=Class1Students("Michael",40)

student2=Class1Students("John",25)
student2.__name="John"
student2.__age=25
student1.speak()
student2.setNameage=("John",25)
student2.speak()



```

	Expected	Got	
✔	my name is Michael, and I am 40 years old. my name is John, and I am 25 years old.	my name is Michael, and I am 40 years old. my name is John, and I am 25 years old.	✔

Passed all tests! ✔

Correct

Marks for this submission: 2.00/2.00.